

Contraction Chamber Geometry and Design

<https://ntrs.nasa.gov/api/citations/19880012661/downloads/19880012661.pdf>

Contraction design for low speed wind tunnels by J James H. Bell and Rabindra D. Mehta:
This paper tested non linear geometry ranging from cubic curvature in the contraction chamber up to 7th order geometry. It provided the impacts of usage of each design for small scale wind tunnels, and also highlighted that based on our scale, orders higher than 7 weren't feasible given the separation requirements as well as ratio of length to height. From their results:

"The contraction wall shape satisfying most of the requirements discussed above is clearly the one given by the 5th order polynomial. It was free of separation (both on the centerline and in the corners) and gave a reasonable Re (~ 400) and flow uniformity (better than 1% variation) at the operating conditions ($U_c = 15 \text{ m/s}$)"

Based on these findings, a 5th order curve was used, with geometry being sourced from the same NASA article. Optimized models predict a bell curved shaped contraction chamber modeled with quintic equation $10t^3 - 15t^4 + 6t^5$

One of the constraints for designing a small scale wind tunnel is manufacturability. Our initial design utilized an 8 by 8 inlet area, but for reasons discussed in the test chamber article, it was necessary to scale up. In order to reach a functional contraction ratio of 4, the dimensions settled upon were 14x14 at the inlet, and 7x7 at the outlet. While it would have been preferred to scale up even more, the complex geometry requires either aluminum folding into this curvature, or 3d printing, which is the more economical option. Below is the math that shows how the CAD model was created based on the quintic equation and provided dimensions.

Width half span inlet: 7, half height inlet: 7, $z=0$

Width half span outlet: 3.5, half height outlet = 3.5 $z=14$

$\Delta x=3.5$ $\Delta y = 3.5$

0.3 In wall thickness

For each corner: top right: $X: 7-3.5(10t^3 - 15t^4 + 6t^5)$ $y: 7 - 3.5(10t^3 - 15t^4 + 6t^5)$
 $z=8t$

Bottom right: $X: 7-3.5(10t^3 - 15t^4 + 6t^5)$ $y: -7 + 3.5(10t^3 - 15t^4 + 6t^5)$
 $z=8t$

Bottom left: $X: -7+3.5(10t^3 - 15t^4 + 6t^5)$ $y: -7 + 3.5(10t^3 - 15t^4 + 6t^5)$
 $z=8t$

Top Left: $-7+3.5(10t^3 - 15t^4 + 6t^5)$ $y: 7 - 3.5(10t^3 - 15t^4 + 6t^5)$
 $z=8t$

$t_1=0$ $t_2=1$

https://cdn.intechopen.com/pdfs/42327/intech-design_methodology_for_a_quick_and_low_cost_wind_tunnel.pdf

Contraction Ratio:

“In the case of wind tunnels for civil or industrial applications, a contractions ratio between 4,0 and 6,0 may be sufficient. With a good design of the shape, the flow turbulence and nonuniformities levels can reach the order of 2,0%, which is acceptable for many applications. Nevertheless, with one screen placed in the settling chamber those levels can be reduced up to 0,5%, which is a very reasonable value even for some aeronautical purposes.”

From this paper, A_i/A_o should be then at least three but optimally 4 or more to contract the flow to provide viable results. Our goal is to reduce turbulence to below approximately 1%, since airfoil measurements require higher precision. As noted, one of the decisions considered is the use of a meshing screen in the settling chamber such that flow is straightened even more beyond the honeycomb straightener. Honeycomb straightener will be discussed in its own article. Also from the article,

“For more demanding aeronautical, when the flow quality must be better than 0,1% in non uniformities of the average speed and longitudinal turbulence level, and better than 0,3% in vertical and lateral turbulence level, a contraction ratio between 8,0 and 9,0 is more desirable. This ratio also allows installing 2 or 3 screens in the set”

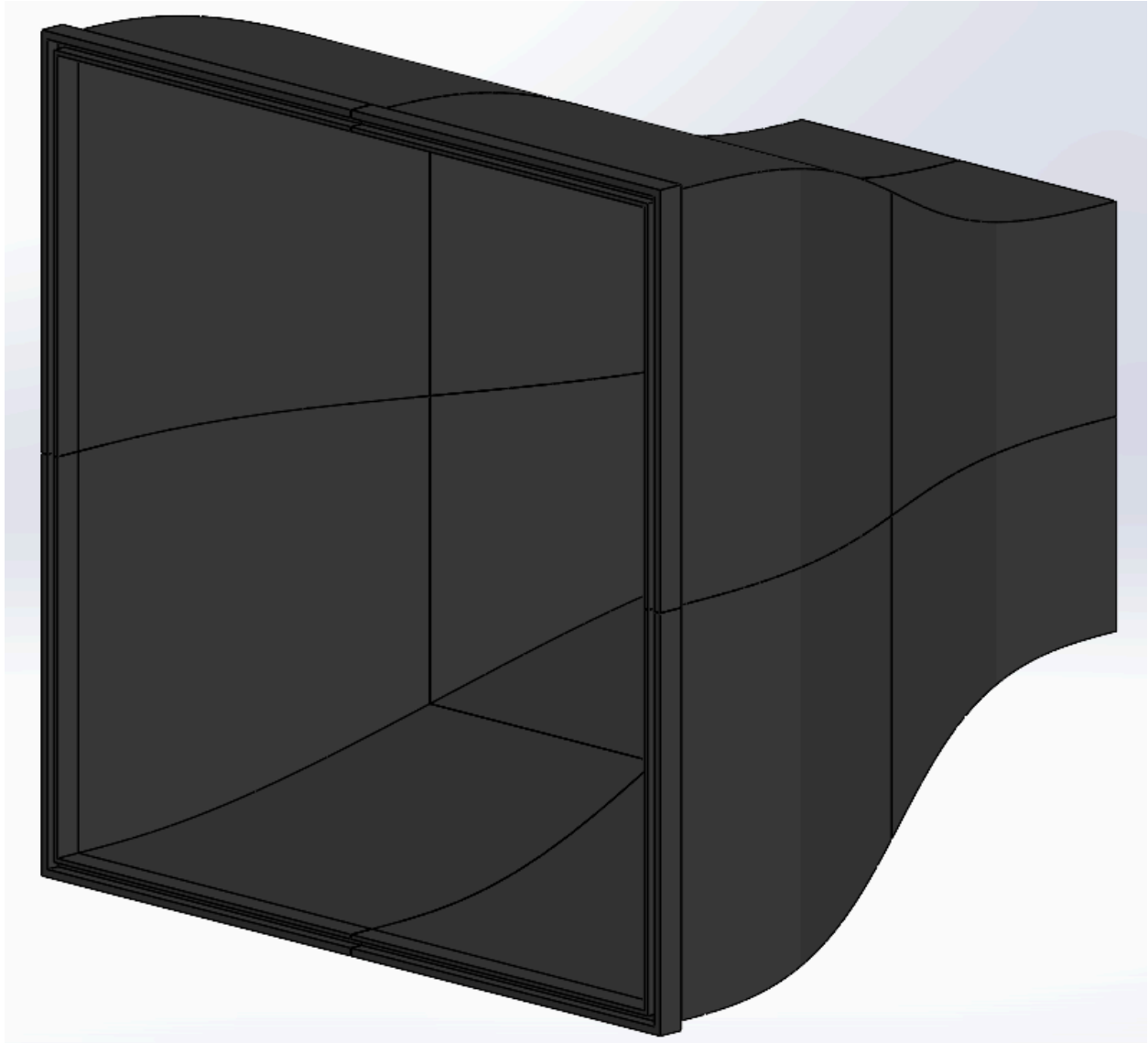
It was necessary to consider the application of our product, as well as the feasibility of production. Given typically 3d printing limits of 10x10x10 chambers, we determined that scaling up to a contraction ratio of 8 would make manufacturing almost impossible without shrinking our test area to an unusable size.

Current design: Our current design utilizes an inlet area of 196 in², an outlet area of 49 in², and therefore a contraction ratio of 4.

Length Ratio: The ratio between the length of the longest outlet dimension of the contraction chamber should be at least 0.8 so as to give the flow ample time to contract and at a more gentle angle. In our case, this ratio is exactly 1. Per Bell and Mehta, their 5th order geometry determined a minimum contraction length of .89

$L_i=14$, $L_z=14$

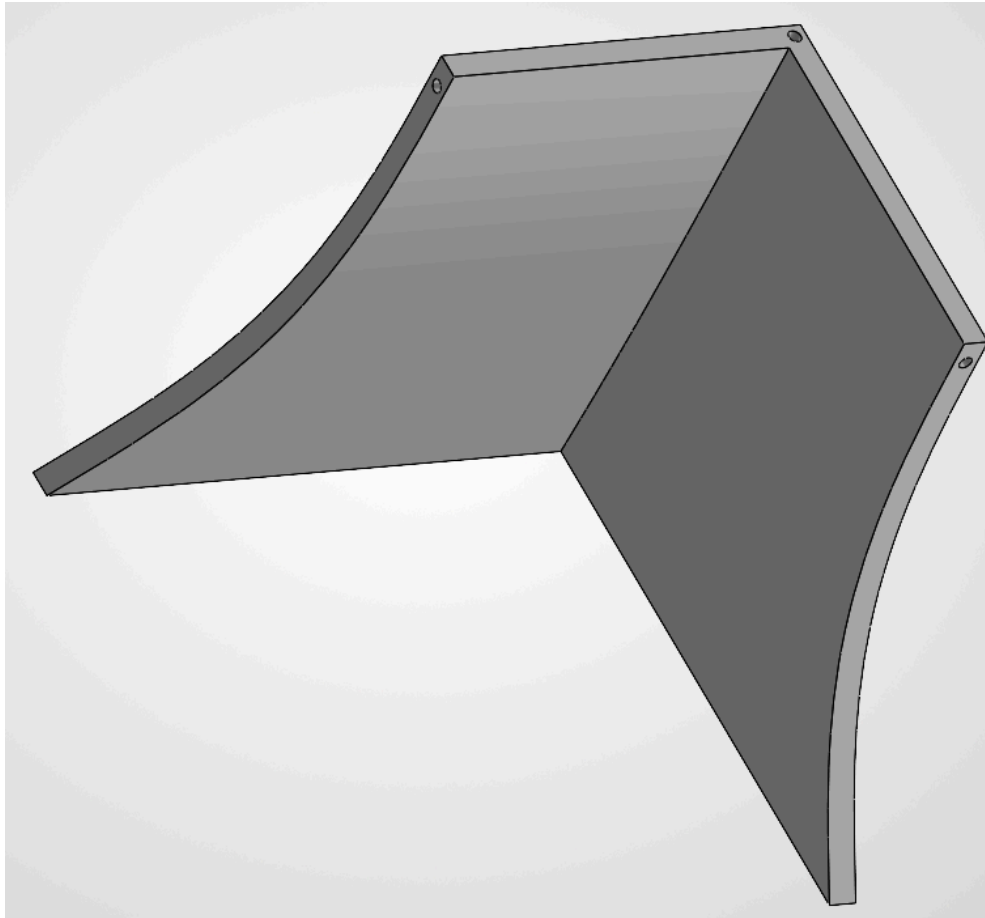
Wind Tunnel Cad Model visualized:



One of the challenges of designing a functional contraction chamber is its attachment to other sections of the windtunnel. It may be difficult to see in the above image, but the inlet of the contraction chamber is slotted such that 14 inch acrylic is capable of fitting in all 4 sides. Each slot is 1/8th inch thick, plus tolerancing of .8 mm, determined by the filament thickness of a 3d printer, and allowing for thermal contraction of the material once fully cooled. The outlet side of the chamber does not use a similar design and instead uses 1/2 inch steel dowel pins, which interface with the test chamber more conveniently, and also provide stabilization of the whole chamber.

Additionally, the chamber is split into 8 sections. This is done for 3d printability, given that each section has dimensions of approximately 7x7x7

View of individual section (1/8th of contraction chamber)



This section corresponds to the outlet facing side, top right from the direction of the inlet. The top flat face with one pin is where this corner attaches to the test chamber. The other two pins are interfaced with other components of the test chamber for alignment and stability. Out of view are similar connection points for the pieces facing the inlet side of the chamber. Because of the curvature of the test chamber, it was necessary to angle pins at the middle of the chamber so that they could align both sections of the chamber.